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A *p*-curve analysis of the emotional Stroop effect among women with eating disorders

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Abstract

Objective: A recent meta-review of attentional bias research in eating disorders suggests that meta-analyses and systematic reviews include many low-quality and underpowered studies (Stott et al., 2021). As such, we examined whether published research examining the link between attentional bias, using the emotional Stroop task, and eating disorders among women with eating disorders has evidential value (ruling out selective reporting of a statistically significant effect) using a *p*-curve analysis. A *p*-curve analysis plots significant *p*-values onto a curve from .01to .05 to examine its distribution. We hypothesized that the *p*-curve would be flat, indicating no true effect. **Method:** The hypothesis, database search strategy, and data analytic approach were pre-registered. The inclusion criteria were reports that compared control and eating disorder groups, reported inferential statistics, and that used body shape/weight or general threat target words. **Results:** Fifty published reports were included in the *p*-curve analyses. Unexpectedly, the half and full *p*-curves were significantly right-skewed, indicating evidential value. However, the results were not robust to the exclusion of the seven lowest *p*-values and on average, reports were underpowered. There were also 18 reports with null results (they had a *p*-value greater than .05), which precluded their inclusion in the *p*-curve analyses. **Discussion:** The findings suggest that most of the evidence from research examining attentional biases using the emotional Stroop task among women with an eating disorder or with elevated eating disorder symptoms is underpowered and so should be interpreted with considerable caution.

Keywords: emotional Stroop; attentional bias; eating disorders; publication bias; p-curve; evidential value; selective reporting; open science; statistical power

A *p*-curve analysis of the emotional Stroop effect among women with eating disorders

There is evidence that people with eating disorders have an attentional bias to stimuli related to their disorder (Stojek et al., 2018; for a meta-review, see Stott et al., 2021). Attentional biases have been measured in different ways in the eating disorders literature, such as with the visual probe task testing attention engagement, the reaction time of the spatial cueing task, the reaction time of identifying words in the visual search paradigm, and the reaction time of the emotional Stroop task. Of the different approaches used to assess attentional bias, the emotional Stroop task is the most frequently used tool in prior research to measure attentional biases among people with eating disorders or with elevated eating disorder symptoms (Stott et al., 2021).

Using the emotional Stroop task, an attentional bias is observed when people take longer to identify the ink colour of disorder-related words (e.g., thin, fat, buff) relative to neutral words (e.g., drill, tool). The attentional bias effect based on the emotional Stroop task has been observed in many conceptual replications comparing women with eating disorders to healthy controls over the past thirty years and is a seemingly robust psychological phenomenon (for reviews, see Brooks et al., 2011; Johansson et al., 2005; Stojek et al., 2018). The observed differences between eating disorder and control groups are used to make inferences about the core psychopathology in eating disorders.

Critically, in a recent meta-review, concerns were raised that the results from attentional bias research in eating disorders may be biased and unreliable because the primary-level studies seem to have low statistical power (Stott et al., 2021). This was particularly the case in the literature that used the emotional Stroop task to examine attentional biases among people with eating disorders (Stott et al., 2021; Eynde et al., 2021). To-date, prior empirical reviews (Brooks et al., 2011; Lavignio et al., 2016) have only

assessed the robustness of the literature using meta-analysis and tests of publication bias. Critically, no prior reviews have quantified the extent of selective reporting in the published literature with statistically significant results and whether the published literature provides evidence of a true underlying effect. The current research addressed this gap in knowledge using a *p*-curve analysis.

P-curve Analysis

There has been growing concern about the rigor and reliability of psychological research in general that stems from many failed replication attempts (Open Science Collaboration, 2015). The failed replications had led some to believe that the psychological literature has low statistical power and that many published findings are false positives (Simmons et al., 2011). Although there are many contributing factors that can help explain failed replications, selective reporting (e.g., manipulated or forced effects) is one agreed upon explanation by may psychology researchers (Baker, 2016). To address the growing concerns about the selective reporting of psychological research, Simonsohn et al. (2014) developed the *p*-curve analysis to help review published reports with statistically significant results.

A *p*-curve analysis is used to empirically examine the evidential value of an existing body of research that reports statistically significant results. Evidential value of a statistically significant effect refers to statistically significant findings that are not due solely to selective reporting (Simonsohn et al., 2014). A *p*-curve analysis plots statistically significant *p*-values from a literature onto a curve from .01 to .05 to examine the distribution of the *p*-values. In the *p*-curve analysis, if an effect is strong and reliable across studies, then the distribution of the *p*-values testing the effect should pool around the lowest end of the *p*-distribution ($p \leq .01$) creating a right-skewed curve, which is indicative of evidential value. In contrast, if there is no effect, then the *p*-curve would be flat, as every *p*-value at or below .05 has a uniform chance of being observed. Likewise, if most *p*-values pool around the cusp of statistical

significance (e.g., .04 and .05) creating a left-skewed curve, then the studies lack evidential value, which suggests the presence of selective reporting in the literature. In this way, the *p*-curve is a powerful tool that can be used to examine whether the published reports with statistically significant results in an existing body of research provides evidence of a true underlying effect.

Of note, *p*-curve analyses have been previously used, in conjunction with meta-analysis, in other eating disorder research, such as the efficiency of different prevention tools for reducing eating disorders in university students (Harrer et al., 2019) and cognitive training to change food-related biases (Navas et al., 2021). Also, Carbine and Larson (2019) examined a *p*-curve for the effect of food-related inhibitory control training. They observed a u-shaped *p*-curve, which suggests that the literature may have evidential value (many *p*-values between .025 and .01), but also that there seems to be some selective reporting in the literature (many *p*-values between .025 and .05). Carbine and Larson (2019) also showed that the evidential value was weak because it was dependent on the inclusion of a single low *p*-value in the analysis. They concluded that the observed effect sizes in the literature were likely biased (inflated) and estimated from low-powered research. In order to improve the reliability of the literature, Carbine and Larson (2019) suggested that researchers should a) pre-register future studies and state an *a priori* hypothesis, b) be more transparent with reporting methods and results, and report on all collected data, c) conduct *a priori* power analyses and d) collaborate across laboratories, when possible, to increase sample size. These examples show the benefit of using *p*-curve analysis for determining the evidential value of a literature, and how results of the *p*-curve analysis can be used to guide future research.

The Present Research

Herein, we examined the evidential value of research that used the emotional Stroop task to assess attentional biases among women with eating disorders. More specifically, we

conducted a *p*-curve analysis of the emotional Stroop research that examined attentional biases among with a diagnosed eating disorder or with eating disorder symptoms. Although any results from the *p*-curve analysis will be informative, we hypothesized that the *p*-curve would indicate a lack of evidential value. The reason is that prior reviews have highlighted the low-quality of eating disorders research that used the emotional Stroop task (e.g., Stott et al., 2021). The current research was pre-registered (plus an amendment) before any data collection or analyses were performed and is available on the Open Science Framework (OSF; https://osf.io/9gmr7/?view_only=e87a6a65c32147828eb5e3b4f78876b5).

Method

Inclusion and exclusion criteria

To be included, reports must have used the emotional Stroop task to test individual differences in reaction times between a sample of healthy individuals (i.e., a control group) and a sample with elevated levels of disordered eating psychopathology or a diagnosed eating disorder. In terms of the emotional Stroop task, reports must have used target words or word/image combinations that were related to body image, food, or general threat (e.g., autonomy threat, ego-self threat, ego-others threat). In addition, to be included, reports should have a clearly stated hypothesis (e.g., "people with elevated eating disorder psychopathology have a slower reaction time than healthy controls on disorder salient stimuli") so as to include the relevant *p*-value in the *p*-curve analyses (see Simonsohn et al., 2014). When examining this criterion, we identified nine reports with ambiguous hypotheses. As such, a decision was made post-hoc to include these nine reports if their literature review and introduction indicated they were testing the hypothesis of interest. As well, reports must have reported sufficient inferential statistics (e.g., the *df* and test statistic of interest) to be included. We also included reports for which enough information was available to infer the *df* for the *p*-value re-

calculation on the *p*-curve application. Moreover, reports needed to be independent of other included reports. Reports not meeting any of these inclusion criteria were excluded.

Selection of Reports

Two database search strategies were conducted to identify primary-level research reports for inclusion into the *p*-curve analyses based on the inclusion criteria. The first database search was conducted to locate systematic reviews and meta-analyses of the literature using PubMed, PsycINFO, and ProQuest databases. We identified search terms related to the emotional Stroop task, eating disorders, and reviews to search the title and abstract of records in each database. The Boolean phrase for each database is provided in Table 1. Results were then filtered for systematic reviews and meta-analyses. The search was conducted on December 21, 2021, and we located 11 reviews. From these located reviews, primary research reports were extracted that fit the inclusion criteria.

The second database search was conducted to locate primary research reports. Like the first search strategy, the reports had to be on the emotional Stroop task among women with eating disorders or elevated eating disorder psychopathology. We also used the same three databases. The second search was conducted on December 21, 2021.

[Include Table 1]

The results of the two searches strategies are reported in Table 2. The pooled sample of research reports that met the inclusion criteria was 50. Of the 50 reports, 32 were included in the *p*-curve (i.e., $p = < .05$ on the hypothesis test of interest). The remaining 18 reports had a *p*-value greater than .05 and so they were not included in the *p*-curve analyses.

[Include Table 2]

Data Extraction

All included reports and their key information (e.g., demographics, extracted test statistic) are reported in Table 2. In line with recommendations for *p*-curve analyses

(Simonsohn et al., 2014), the following information was extracted from each report and included in the extended disclosure tables on OSF (see "Included Studies Extended Disclosure Table" for the extended disclosure table of all included articles and see "P-curve Full Disclosure Table (All Studies)" for all reports screened, including excluded reports and rationale for exclusion): 1) quoted text of the predicted hypothesis; 2) study design; 3) the number of participants and simple demographics (e.g., gender, age, how eating disorder psychopathology was measured); and 4) quoted text with test statistics. In addition, we collected information on how the emotional Stroop task was designed (e.g., type of target and control words, whether images were included) as well as how interference was measured. In cases where a combination of multiple different target words (e.g., body shape/weight and food words) were examined separately, the first *p*-value of interest to appear in the paper was extracted. In these cases, a secondary robustness test was conducted based on the second *p*-value testing the main hypothesis for that category. However, only 18 reports met the criteria for a robustness test, of which seven did not reach statistical significance. Because only 11 reports were included in the robustness *p*-curve, it is difficult to reliably interpret these results, so they were not reported.

Most reports provided the necessary information for the *p*-curve shiny app to calculate the exact *p*-value. When information was missing from a given report, we contacted the author(s) of these reports to provide us with needed information. This was the case for three reports. Unfortunately, none of the authors responded and so these reports were excluded from the analyses. There were also three reports with findings that were not statistically significant and that did not report inferential statistics. These three reports were still included in the disclosure table and considered in interpretations of findings in the Discussion section despite not being included in the *p*-curve analysis. Hence, the number of reports included in Figure 1 is 47, but 50 reports were considered in the current review.

Data Analytic Procedure

All extracted *p*-values (*n*= 47, as three non-significant tests were not reported and could not be calculated) were analyzed using the online *p*-curve app (see: http://www.p-curve.com/app4/). The *p*-curve app runs two analyses to determine evidential value. The first analysis is a binomial test for both right skew (i.e., testing if most reports fall at the lower end of the curve, such a $p < .01$, indicating evidential value) and flatter than the 33% power curve (i.e., testing if the *p*-curve is flatter than one would expect if the reports had 33% power; if this test is significant, it suggests lack of evidential value). The binomial test compares the number of *p*-values in the lower bin, defined as values below $p = .025$, to the *p*-values in the upper bin, defined as values above $p = .025$ (Simonsohn et al., 2014).

The second analysis is a full continuous test that examines the whole curve rather than binning the results. It includes all *p*-values and creates *z*-scores of each score's probability of having its observed value between 0 and .05, if the null hypothesis was true (Simonsohn et al., 2015). To aggregate these values, the *p*-curve app divides the sum of the *z*-scores by the square root of the number of tests included in the analysis.

Additionally, a robustness test is run to examine how the curve (both the skewness and 33% power) is affected by removing the lowest or the highest *p*-values, which allows for identifying extreme values affecting the *p*-curve (Simonsohn et al., 2015). One can be more confident in sets of reports wherein evidential value is maintained after eliminating the most extreme results.

Lastly, the *p*-curve app generates a diagnostic plot for the power estimate of included reports. The estimated power and diagnostic plot are created by calculating the Stouffer *z*-scores at each power level between 5% to 90%. The estimated power across reports will be the point with the best fit (i.e., smallest *z*-score for the Stouffer test).

Results

Descriptive analyses

The 50 research reports included 3,996 participants (M = 80, SD = 50.55), with a minimum of 12 participants and a maximum of 321 participants. Fourteen reports provided the ethnicity of their participants. All demographic information associated with each study is reported in Table 2. Of the 50 included research reports, only 14 reported sample demographic information. In these 14 reports, all involved participants that were Caucasian, but the number of Caucasian participants varied across the 14 research reports (range: 52–100%). Furthermore, of the 14 reports, seven recruited Asian participants. Asian participants in these reports ranged between 0.7 – 29%. Similarly, six of the 14 reports included African American or Black participants. The number of African American or Black participants in these reports ranged from 1– 34%. Likewise, five of the 14 reports included participants that did not identify with provided racial categories (range: 2–22%). Moreover, four of the 14 reports included Latinx participants (range: 1 – 17%). Lastly, of the 14 reports two included few participants that were Native American (range: 1.8–3.3%) and two reports included some participants that indicated more than one race or ethnicity (range: 6.5-9%).

As for participants' sex, all 50 reports included female participants. The range of female participants ranged from 55 – 100%. However, only six of the 50 reviewed reports included male participants. Among these six reports, male participants in the samples ranged from 14 – 45%.

Only five reports included information on socioeconomic status of participants, and each of these reports measured socioeconomic status using different tools. Among these five reports, socioeconomic status was described as "wide" (Ben-Tovim & Walker, 1991), "middle" (Green et al., 1999), and among "the highest three SES classes" (Wilson, 1999). One more specific study reported the majority (i.e., 56%) of participants made < $20, 000 annually, 8.3% made $20,000 – $50,000 annually, 18.3% made $50,000 - $100,000 annually,

and 6.7% made > $100.000 annually among a pool of undergraduates (Lokken, 2001). In two reports, most participants had at least university-entrance levels of education (Preuss et al., 2019; Redgrave et al., 2008). Of the fifty reviewed reports, thirty-two used university students (e.g., undergraduate, medical) as the control group or entire participant pool.

P-Curve Analysis

Figure 2 depicts the visual of the *p*-curve analysis (see the solid line for observed *p*-curve results). See Table 2 for information on each study. Of the 32 statistically significant reports included in the *p*-curve analysis, 24 *p*-values testing the key hypothesis were smaller than .025.

To first examine the right-skewness of the *p*-curve, results for the binomial test, $p = .004$, the full continuous test, $z = -5.67$, $p < .0001$, and the half continuous test, $z = -4.65$, $p < .0001$, were all statistically significant (see Figure 2). The *p*-curve was also not significantly flatter than the 33% power curve (the dashed curve in Figure 2), binomial: $p = .774$; half: $z = 6.85$, $p > .999$; full: $z = 1.81$, $p = .965$. Together, these results indicate evidential value in the literature (i.e., a right-skewed distribution).

[Insert Figure 2 – p-curve]

Importantly, however, the estimated power across the reports with statistically significant findings was 57%, 90% Confidence Interval = [36%, 75%], as identified by the diagnostic plot in Figure 3. If the same reports were all run again at 57% power, half of the time, the *p*-curve would be flatter than 33% power curve (see dashed line in Figure 2) and the other half of the time it would be right-skewed.

[Insert Figure 3 – power estimate]

We tested the robustness of the findings when excluding the most extreme *p*-values for both the test of right skewness and the flatter than 33% power curve test. Critically, the full *p*-curve for right skewness loses evidential value after removing the seven lowest *p*-

values (i.e., close to a quarter of the included reports; see Figure 4a). Likewise, the half *p*-curve for right skewness loses evidential value after removing just three of the lowest *p*-values (Figure 4b). Thus, excluding a few of the lowest *p*-values eliminates evidential value. In contrast, the full *p*-curve was not statistically flatter from the 33% power curve even after removing upwards of 15 reports with the lowest *p*-values (see Figure 4c), which suggests the presence of evidential value.

[Insert Figure 4 – robustness tests]

Discussion

We conducted a review of 50 emotional Stroop reports among women with eating disorders symptoms or a diagnosed eating disorder. Of the 50 reports, 47 were able to be included in the *p*-curve analyses to examine the evidential value of the literature. Based on Stott and colleagues' (2021) meta-review of the attentional bias literature in eating disorders, we hypothesized that the *p*-curve would be indicative of no evidential value (i.e., flat). Unexpectedly, the *p*-curve had a significantly right-skewed distribution, which indicates evidential value (Simonsohn et al., 2015).

Critically, however, other results from the *p*-curve analyses suggests that evidential value was weak as the robustness test indicated that the evidential value was dependent on the inclusion of a few low *p*-values. In particular, the full *p*-curve loses evidential value after removing seven of the lowest *p*-values. Likewise, the half *p*-curve loses evidential value after removing only three of the lowest *p*-values. Taken together, the results of the *p*-curve analyses indicate that results from the literature examining attentional biases among women with eating disorders using the emotional Stroop task should be interpreted with considerable caution.

Of import, the estimated power across studies with statistically significant results in the *p*-curve analysis was below 80%—the minimally acceptable level of power in

psychological research. In addition, even though the *p*-curve was quantitatively different from the 33% power curve (see the dashed line in Figure 2), the visual shape of the *p*-curve resembled the shape of a 33% power curve. Specifically, the *p*-curve, depicted in Figure 2 as the solid line, has an increased number of *p*-values at the .01, .03, and .05 levels. The *p*-values at .02 and .04 fall below the dashed power line. The *p*-curve begins to appear flatter than the 33% curve after removing the six lowest *p*-values, where the 33% power analysis becomes approximately $p = .10$. These findings indicate that many of the included reports likely have less than 33% power, which is woefully below the traditional minimum of 80% power used in psychology research.

It is well-established that results from low-power research are unreliable because they have a low chance of detecting a true effect (Cohen & Rogers, 2003). They may also not reflect a true effect (e.g., Button et al., 2013). For these reasons, results from low-power research are almost always inconclusive and so should be interpreted with considerable caution. As such, although there is evidential value based on the *p*-curve analysis in the current research, many of the analyzed reports have low power, which undermines confidence in the results of the literature examining attentional biases among women with eating disorders using the emotional Stroop task.

Another concern with the reports included in the *p*-curve analysis is that some did not have a clear hypothesis. Simonsohn et al. (2015) advised that only *p*-values directly relevant to the stated hypothesis should be included in the *p*-curve analysis. The reason is that the accuracy and reliability of the null-hypothesis significance test rely on clear, theory-driven, *a priori* hypotheses. Critically, when extracting information from the included reports in the *p*-curve analyses, nine reports did not clearly state a theory-driven hypothesis (reports that found a statistically significant effect: Ben-Tovim et al., 1989; Channon et al., 1988; Cooper & Fairburn, 1993; Davidson & Wright, 2002; Green et al., 1994; Kanakam et al., 2013;

reports that did not find a statistically significant effect: Preuss et al., 2019; Tapper et al., 2008; Tressler, 2008). That said, we made the decision post-hoc to include these nine reports to ensure that we included as much as possible of the relevant literature in the *p*-curve analyses. Of note, in an exploratory analysis, the results remained virtually the same after excluding these nine reports (see OSF for these analyses).

We are also concerned with reports on attentional biases among women with eating disorders using the emotional Stroop task that were not included in the *p*-curve analyses. More specifically, there were 18 reports that had a *p*-value greater than .05 (Green & Rogers, 1993; Hodge et al., 2020; Jansen et al., 1998; Kuehfuss, 2007; Lattimore et al., 2000; Lee et al., 2017; Markis, 2015; McManus et al., 1996; Ogden & Greville, 1993; Perpiñá et al., 1998; Preuss et al., 2019; Prystanski, 2011; Quinton, 1998; Scholar, 1999; Seddon & Waller, 2000; Tapper et al., 2008; Tressler, 2008; Wilson, 1999), which precluded their inclusion in the *p*-curve analyses. When juxtaposed with the 32 reports included in the *p*-curve analysis, the fact that there are at least 18 reports (36% of the reports that fit our inclusion criteria) with results that are not statistically significant is indicative of contradictions in the published literature. Critically, the contradictions in the literature may be due, in part, to low power (Maxwell, 2004). Unfortunately, the *p*-curve application does not consider the non-significant results when generating the power estimate, so we do not know if the results from reports that are not statistically significant are adequately powered or not. As such, the results indicating evidential value from the *p*-curve analyses should be interpreted considering the substantial number of reports with results that are not statistically significant.

It is possible that methodological differences may exist between reports with statistically significant results and reports with results that were not statistically significant. For example, Tabri and Palmer (2020) noted word repetition (people are faster at recognizing words they have already seen) and the use of unmatched words on different lexical

characteristics (e.g., number of letters, syllables, orthographic neighbourhoods, and frequency of usage in the English language) as limitations in the eating disorders emotional Stroop literature, among others. These differences in the task and nature of the stimuli may help explain differences between reports with and without statistically significant results. For instance, among the statistically significant reports reviewed herein, there was more variance in the type of word, or word and image stimuli compared to reports that found no statistically significant difference between groups. As well, some non-statistically significant reports only used food words and control words or body shape/weight words and control words, compared to the statistically significant reports which more commonly combined the two groups of target words (i.e., had both food and body words). It is possible that these methodological differences may help explain why some reports have statistically significant results and other do not. Future research should examine methodological differences between reports with statistically significant findings and reports with results that are not statistically significant in a more rigorous and systematic approach.

The findings of the current research also have implications for interpreting the results of prior reviews on the emotional Stroop task among women with eating disorders. Of note, we identified 11 meta-analytic reviews (see Figure 1) from which we included many of the reports in the *p*-curve analyses. Critically, like the included reports, the findings from these reviews should be interpreted with considerable caution. The reason is that the inclusion of low-powered reports in the prior reviews will produce conclusions that are unreliable and inaccurate (e.g., Aguinis et al., 2011).

In sum, the results of the *p*-curve analyses and the fact that there are many published reports with results that are not statistically significant should give everyone pause about the reliability and accuracy of results based on the emotional Stroop task in the eating disorders literature. Accordingly, there is a need for high-powered (i.e., at least 80%; Bakker et al.,

2016) pre-registered research. Often, it is difficult to determine the effect size for *a priori* power analyses and so researchers can consider initially conducting a hypothesis-generating or exploratory study followed by a high-powered replication study that confirms the results (see more information about sample size justification and *a priori* power analyses in Lakens, 2022). The research conducted by Tabri and Palmer (2020) using the emotional Stroop task in the context of eating disorders provides an example of this approach with pre-registration.

A possible limitation of the current research is that we only evaluated published research that provided the necessary inferential statistics for the *p*-curve analysis. As such, there may be unpublished research that could have been included in the *p*-curve analysis. However, it is likely that many of these unpublished reports have results that were not statistically significant, which is consistent with the notion of publication bias. Accordingly, the inconsistent findings observed in the reviewed literature should be interpreted with the understanding that there are likely other reports that were not published because they had results that were not statistically significant.

Another limitation has to do with the *p*-curve analysis approach. Results of a recent simulation study showed that the *p*-curve analysis overestimates power when there is heterogeneity in effect and sample sizes among the included reports (Brunner & Schimmack, 2020). As such, to the extent there are differences in sample size and effect sizes across the included reports, the 57% power estimated across reports included in the *p*-curve analysis may be an overestimation. This means that the overall estimated power may be even lower.

Furthermore, from our review, we observed that emotional Stroop research in context of eating disorders may have limited external validity. Most reports involved women with eating disorders and so it is unclear whether findings generalize to populations of men with eating disorders. Likewise, there was a tendency for reports to rely on samples of undergraduate women as both the control group and the group with elevated disordered

eating characteristics. With this approach, it is unclear whether findings would generalize to populations who do not have access to post-secondary education. As well, there was limited information on the socioeconomic status of participants in the reports, but the use of university participants suggest middle to upper-class participants in these samples. Moreover, most reports recruited participants residing in "Western" countries (e.g., Canada, USA, and the UK), except for two reports that involved samples from Iran (n = 98) and Korea (n = 39). Taken together, the external validity of results to men, people without access to post-secondary education, and to people residing in non-Western countries remains unresolved.

Conclusions

Concerns have been raised about the low power of results from research examining attentional biases among women with eating disorders using the emotional Stroop task (e.g., Stott et al., 2021; Van den Eynde et al., 2011). Herein, we used *p*-curve analyses to empirically determine the evidential value of the literature reporting statistically significant results. Although results suggest the presence of evidential value, a robustness check indicated that the evidential value was dependent on the inclusion of few small *p*-values. Moreover, most of the included reports had low power and results from low-powered research are inconclusive. There was also a substantial number of reports with results that were not statistically significant, which precluded their inclusion in the *p*-curve analyses. This pattern of contradictions in the literature is consistent with the literature being low-powered. Accordingly, results from research examining attentional biases among women with eating disorders using the emotional Stroop task should be interpreted with considerable caution.

Public Significance Statement

Concerns have been raised about the low quality of research examining attentional biases among women with eating disorders using the emotional Stroop task. In the current research, we observed that the evidential value of primary research reporting differences between women with and without eating disorders was equivocal and had low statistical power. These results can guide researchers towards conducting more rigorous research on attentional biases among people with eating disorders.

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References marked with an asterisk indicate reports included in the *p*-curve analysis

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Figure 1

Prisma flow chart for the original and amended pre-registration demonstrating the articles included in the analysis from meta-analyses and systematic reviews.

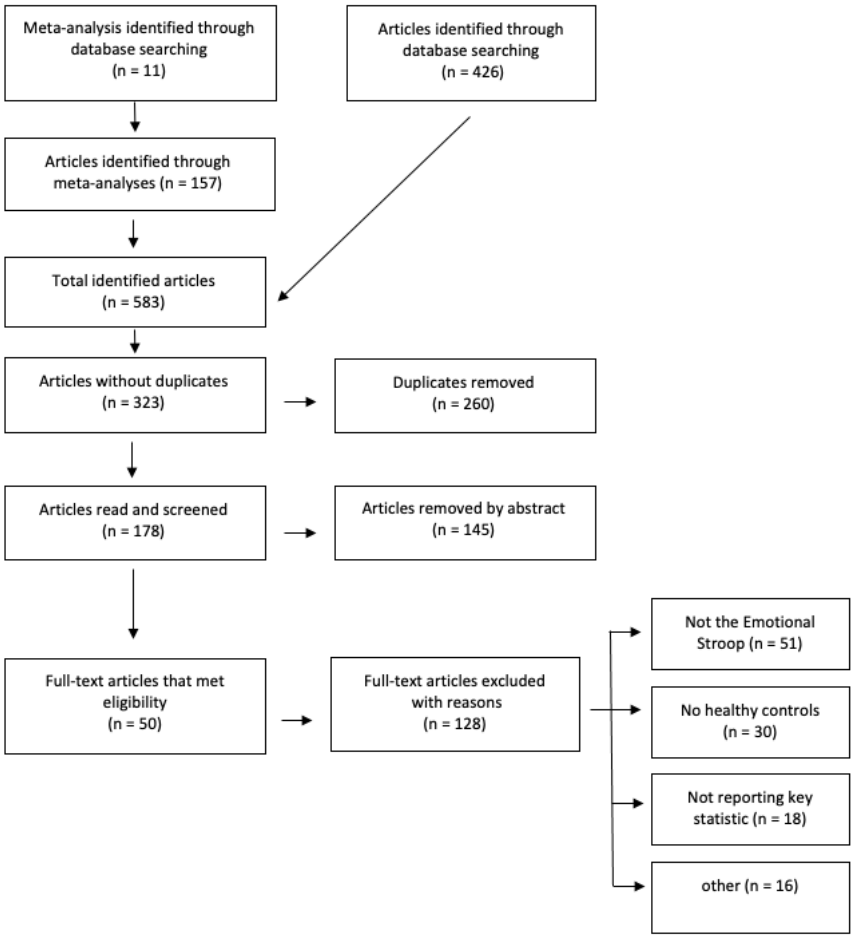
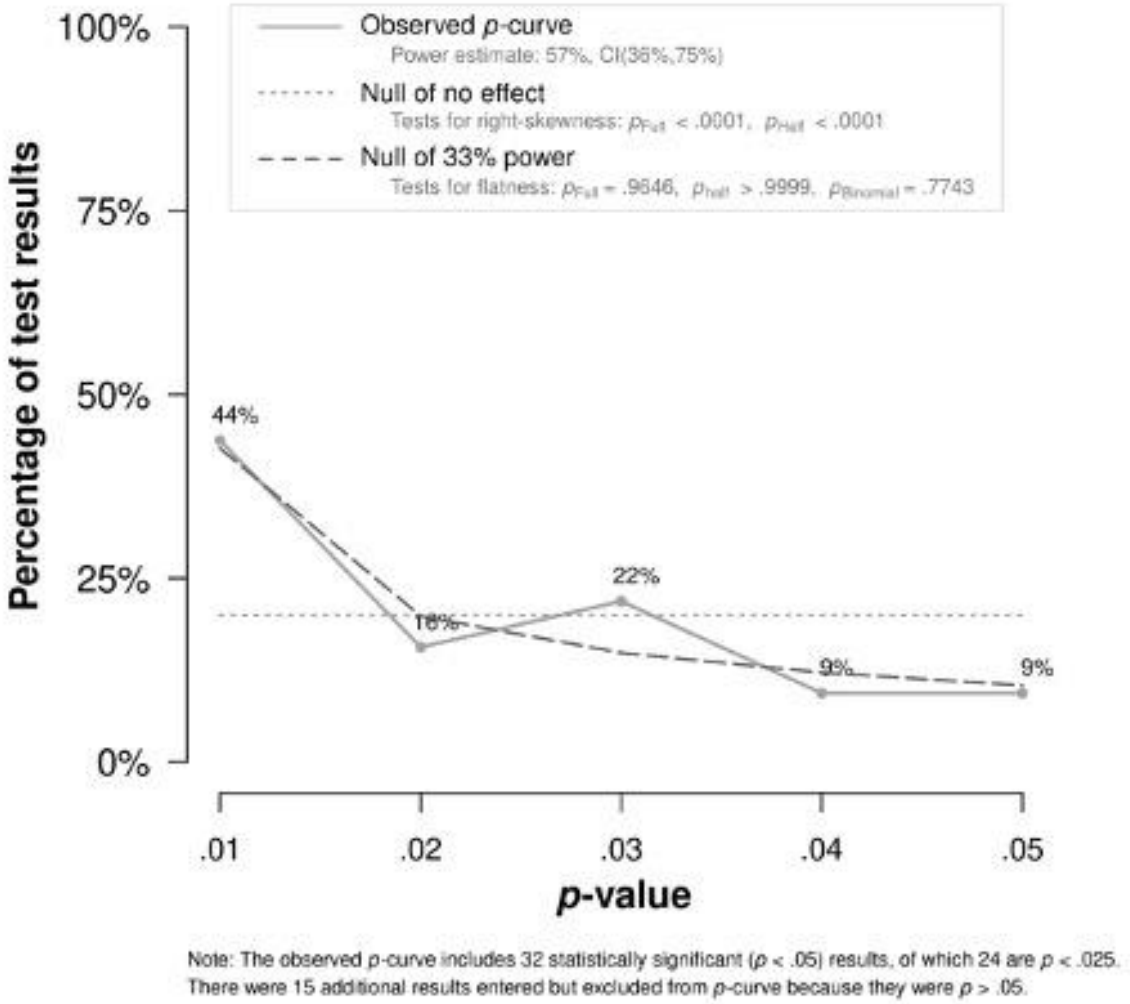


Figure 2

A statistically significant right-skewed *p*-curve for the set of included reports (n = 50), indicating that these reports contain evidential value.



Note. The observed *p*-curve includes 32 statistically significant (p < .05) results, of which 24 are p <.025. An additional 15 reports had p >.05 for the statistic of interest. This figure was generated by *p*-curve app 4.06.

Figure 3

A diagnostic plot of power. This plot shows the best fitting level of power by examining the stability of the computed p-curve at different values of power between 5% and 99%. The best fitted power of this p-curve is 57%.

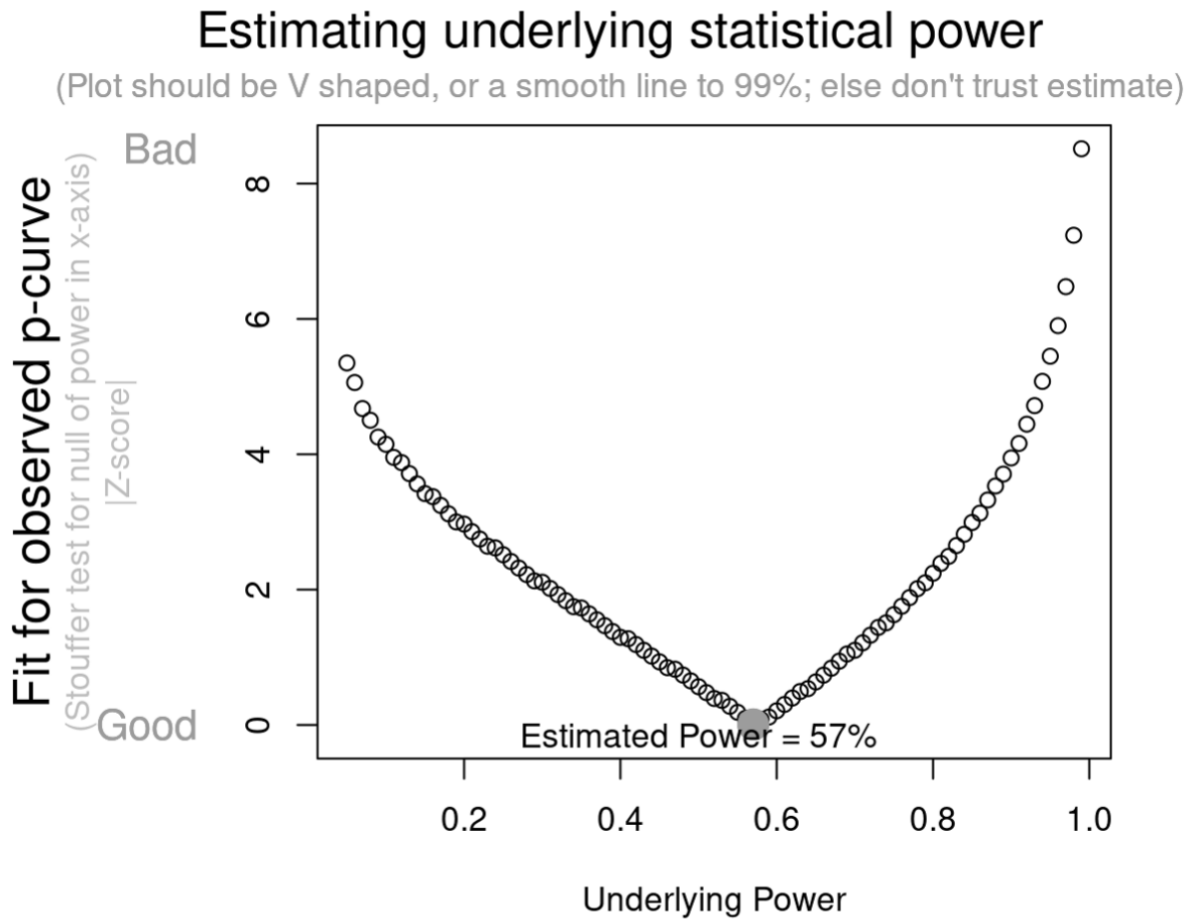
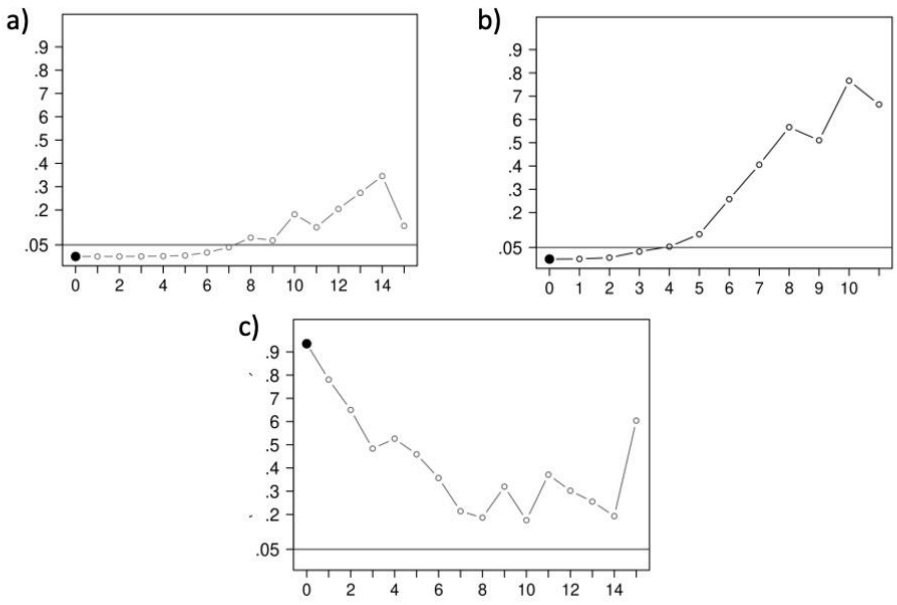


Figure 4

These graphs depict the effect of dropping the lowest *p*-values (the x-axis) on the significance (y-axis) of the full *p*-curve test for right-skewness (a), the half *p*-curve test for right-skewness (b), and the test for flatness relative to 33% power (c).



Note. Within each graph, the solid horizontal line reflects the significance threshold (p = .05), and the dotted line is the result reported in the text. This figure was generated by *p*-curve app 4.0

Table 1.

Boolean phrases for the search strategy by databases.

Databases	Emotional Stroop	Eating disorder
PsychInfo	(ab(Stroop) OR ab(attentional bias) OR ti(Stroop) OR ti(attentional bias)	(ab(eating disorder*) OR ab(purging) OR ab(anorexi*) OR ab(bulimi*) OR ab(binge eating) OR ti(eating disorder*) OR ti(purging) OR ti(anorexi*) OR ti(bulimi*) OR ti(binge eating)
PubMed	(stroop[Title/Abstract] OR attentional bias[Title/Abstract])	(eating disorder[Title/Abstract] OR feeding and eating disorders [Title/Abstract] OR anorexia [Title/Abstract] OR bulimia [Title/Abstract] OR binge eating disorder [Title/Abstract])
ProQuest	(ab(stroop) OR ab(attentional bias) OR ti(Stroop) OR ti(attentional bias))	(ab(eating disorder*) OR ab(purging) OR ab(anorexi*) OR ab(bulimi*) OR ab(binge eating) OR ti(eating disorder*) OR ti(purging) OR ti(anorexi*) OR ti(bulimi*) OR ti(binge eating))

Note. The Boolean phrases for emotional Stroop and eating disorder in each database were combined using the AND operator.

Table 2.

Disclosure table for included reports.

Paper	N	Measure of disordered eating and Grouping Strategy	Additional Demographic Info.	Study Design	Key Statistical Result	Results	Classification	How was interference measured?	Notes	True p-value
Abraham (2007)	76 female undergraduate students living in the US (40 high BUILT-R, 36 Low BUILT-R)	Non-clinical group. High BUILT-R scores vs. Low BUILT-R scores.	Mage 19.5; 89.2% Caucasian, 1.8% Native American; 1.8% Asian; 1.8% Hispanic; 5.4% African American; Average BMI 22.44.	2 (BN Symptoms: High BUILT-R, low BUILT-R) X 4 (Word type: good, body shape, interpersonal, and neutral) Repeated ANOVA.	Two-way attenuated interaction (BN symptoms X word type)	*F*(3, 74) = 3.72	Food, body shape, interpersonal, and neutral control	RT (milliseconds)		0.01501

Albery et al (2016)	88 females (45 BN, 43 CT) living in England.	BN group was recruited through 12-step fellowships in the community of females who self-identified as having BN. The CT group was recruited from a London university.	Mage 30.4. Mbmi for BN group 28.9; Mbmi CT 31.9.	3 (word type: food, body, neutral) X 2 (group: BN, healthy control) ANOVA	t-test (interference score difference between BN and control on food-related word)	*t*(86) = 3.406	Food; robust: body-related	RT (milliseconds)	0.001
Ben-Tovim & Walker (1991)	115 females in Australia; 22 AN, 27 BN, 29 CT with high	females with AN and BN were drawn from treatment centres, control	Mbmi = AN 24.4; BN 26.8; high DFT 13.6; low DFT 14. Controls were drawn from a private school "drawing individuals from a wide	4 (group: AN, BN, control high weight concern, control low weight concern) X 5 (Stroop Levels: circles, true stroop, food, food control, body shape, body	One-Way Main effect of Group for Food Words	*F*(3,111)= 4.35	Food	RT difference between Target and Neutral	0.00615

	DFT, 37 CT with low DFT	group was from private schools. All participants completed measures of Drive for Thinness (i.e., high vs. low).	socioeconomic base".	shape control) ANOVA				Word (Seconds)		
Ben-Tovim et al (1989)	N = 74 females in Australia; 17 with clinical diagnosis of AN; 19 with BN; 38 CT.	Participants were recruited from in-patient Tx centre for eating disorders. Control participants were staff and medical students from the clinic.	AN group Mage 25; BN group Mage 27; CT group Mage 23.	4 (group: AN, BN, control high weight concern, control low weight concern) X 6 (card type: circles, true stroop, food, food control, body shape, body shape control) ANOVA	Two-way attenuated interaction (group X card type)	*F*(2,73)=4.67	body, food	RT (seconds) to name the ink colour are transformed into disruption score: The Disru	Unclear hypothesis. Interaction not included. Recorded main effects here.	0.01234

ption scores were obtained by subtracting control from target color naming times (e.g., Stroop disruption = True Stroop - circles; shape disruption =

								shape words - shape control words)		
Black et al (1997)	N = 45 females in the USA; 15 with BN; 29 CT (16 restrained eating; 13 unrestrained eating).	Participants with BN were receiving outpatient Tx at an ED clinic. Control group was university students.	BN group Mage 23.8, Mbmi 23; CT group with restrained eating Mage 20.6, Mbmi 23.7; CT group with unrestrained eating Mage 21.9, Mbmi 21.2.	3 (group: BN, restrained, unrestrained) X 4 (words types: colour, control, food, weight/shape) ANOVA	Two-way attenuated interaction (group X words type)	*F*(2, 42) = 3.74	food, weight/ shape	RT (seconds) transformed into a discrepancy score.	Interaction not reported. Main effect reported here.	0.032

Braet & Crombez (2003)	74 children in Belgium. 34 with elevated DEB-Q (24 girls, 10 boys) and 40 CT from a school (29 girls, 11 boys),	Participants in the ED group were receving in-patient Tx for severe obesity and had elevated DEB-Q scores. Controls were recruited from schools.	Mage 14 years. Children in Obes. Group were on average 77% above the ideal weight for their height. The control group was on average 3% above the ideal weight for their height.	2 (Group: Children with obesity vs. Control) X 2 (Word Presentation Condition: Blocked vs.Mixed) X 2 (Word Type: Experimalestal vs. Control) X 2 (Word Category: Food vs. Negative Emotion) Repeated measures ANOVA	Two-way attinuated interacton (group X word category)	*F*(1,72) = 10.00	Food vs Negative meotion words	RT (milliseconds)	The study was examining a 4-way ANOVA, for our purposes, only the two-way ANOVA between group type (children with obesity vs CT) by Word category	0.00229

									was extracted.	
Channon et al (1988)	N = 40 females in the UK. 20 AN in out-patient Tx program, 20 CT.	Group with AN met Russell's (1970) diagnosis criteria.	AN group Mbmi 12.5; CT Mbmi 21.1)	2 (Group: AN vs healthy control) x 2 (Word type: food vs control) ANOVA	Two-way attenuated interaction (group X words type)	*F*(1, 38) = 13.8	Food	Not reported in the paper	No clear hypothesis.	0.00065
Cooper & Fairburn (1992)	N = 60 females in the UK; 12 AN, 12 BN 12 CT, 12 dieters with ED symptoms, 12 dieters	Participants were receiving outpatient Tx for eating disorders. Controls were recruited by		5 (Group: AN, BN, non-symptomatic dieters, symptomatic dieters, CT) X 2 (Card type: target [body shape/weight/food], control) ANOVA	Two-way attenuated interaction (Group X Card type)	*F*(4,55)= 3.06	mixed words - food, weight/ shape	RT (seconds)	Unclear hypothesis, otherwise correct.	0.02392

	without ED sypmtoms.	advertisement. ED symptoms among the CT group was measured using the EAT.							
Cooper et al (1992)	N = 54 females in the UK: 36 in outpatient Tx for BN; 18 CT	Clinical diagnosis (DSM-II-R). Controls were recruited by placing signs looking for female participants to complete a study on eating, shape, and weight.	BN group Mage 24, Mbmi 21.8; CT group Mage 22.1, Mbmi 3.5	2 (Group: bulmia, healthy control) X 2 (Interference index: target vs. control words, colour vs. control words) ANOVA	Two-way attenuated interaction (Group X Interference index)	*F*(1, 52) = 5.67	mixed words - food, weight/ shape	Not reported in the paper	0.02095

Davidson & Wright (2002)	N = 66 females in the UK; 20 in outpatient Tx for BN; 18 age-matched CT; 18 button-pressing CT; 18 voice-activated CT.	Clinical diagnosis (DSM-II-R). CT group was university students and staff.	BN group Mage 25.5, Mbmi 21.2; CT group Mage 24.9, Mbmi 21.1; button-pressing CT Mage 19.7, Mbmi 22; voice-activated CT Mage 19.1, Mbmi 21.1.	2 (Group: BN, healthy controls) X 3 (Disruption score: colour, food, shape/weight) ANOVA	Two-way attenuated interaction (Group X Disruption Score)	*F*(1,33)=4.53	mixed words - weight/ shape and food	RT in hundredths of a second. Disruption scores represent the time taken to colour name the control words subtracted from the time taken to colour	Unclear hypothesis, otherwise correct. Participants with BN were slower on control words as well.	0.04086

								name the target words.	
Drier et al (2021)	N = 321 females living in USA. 139 elevated EDEQ; 122 CT; 60 with peanut allergies.	Elevated EDEQ vs. non-elevated.	Mage 5.4. Mbmi elevated EDEQ = 29.86; CT = 24.44; peanut allergies = 23.65). Primarily (80%) caucasian, 0.7% Asian, 7.9% African american/Black, 5% Hispanic/Latinx, 6.5% more than one race/ethnicity.	3 (Group: elevated EDEQ score, peanut allergy, healthy control) X 6 (Stroop words: food, body, peanut, positive, negative, neutral)	Two-way attenuated interaction (Disordered eating X Stroop condition)	F(8, 311) =5.12	food, body	RT (milliseconds)	0.00001
Ewald (1993)	N = 69 females living in the US. 18 AN; 15 BN; 18 with social phobia; 18 CT	The ED group was recruited from an outpatient Tx program. Participants with	Group with AN (Mage = 25.17), group with BN (Mage = 24.40), Group with social phobia (Mage. = 29.94), CT (Mage = 24.94). 20 Asian, 36 Caucasian, 1	4 (Group: AN, BN, social phobia, CT) X 3 (Word Type: General Negative Evaluation, Negative Evaluation by others, negative shape) Repeated measure ANOVA	Two-way attenuated interaction (Group X Word type)	*F*(6, 126) = 2.51	Negative Shape words, Negative Evaluation words, General	Interference indices: the response latency of the	0.02501

		social phobia were recruited from outpatient Tx centres for anxiety disorders. CT participants were recruited from undergraduate psychology courses.	Hawaiian, 6 Mixed Race, 4 Other.				negative	control card was subtracted from that of the target card (in seconds).	
Fairburn et al (1991)	N = 74 females in the UK; 24 BN, 50 healthy controls.	Participants with BN had a diagnosis from a medical practitioner and were receiving Tx.	Males also participated, N = 24 for another unrelated analysis.BN group Mage 21.3, Mbmi 22.6; CT group Mage 20, Mbmi 20.9.	2 (Group: BN, healthy controls) X 3 (Card type: colour, target[food/shape], control) ANOVA	Two-way attenuated interaction (Group X Card type)	*F*(2,144)= 5.6	mixed wors - weight/ shape and food	time taken (seconds) to complete a card with 100	0.00455

		Control participants were university undergraduate students. All participants also completed the EAT.						words on it		
Fassino et al (2002)	N = 40 females in Italy; 20 AN; 20 CT.	AN group were receiving Tx for AN and had a DSM-IV diagnosis. CT were recruited from high school and university classes.	AN group Mage 23.8, Mbmi 15.16; CT group Mage 23.1, Mbmi 20.60.	Two cell: AN vs CT	Mean differences	*t*(38) = 2.03	Food, Weight/ shape with negative overtones	Errors in Stroop Effects	We calculated the t values from the mean, sd, and sample sizes from the paper.	0.0494

Francis et al (1997)	N = 28 undergraduate females in Canada. 14 CT, 14 elevated RS score	RS: Restraint group *M* 21.43; Unrestraint group *M* 8.57.	Mbmi = 22.43	2 (Group: restrictive eaters vs non-restrictive eaters) X 2 (Block: 1 vs 2) X 2 (Food word: unrestrictive vs restrcitive) ANOVA	Main effect of Group	*F*(1, 26) = 6.50	Food	Latency difference between Normal Stroop task and Food related Stroop Reaction Time (milliseconds)	0.01703
Green & Rogers (1993)	N = 55 undergraduate females in the UK; 27 low DEB-Q, 15 high DEB-Q,	Low DEB-Q *M* 2.19; High DEB-Q *M* 3.62; Dieters *M* 3.7	Age range 18-40;	2 (Word: food vs body shape) X 3 (Group: high restraint, low restraint, active dieting) ANOVA	Two-way attenuated interaction	*F*(2, 52) = 2.95	Food & Body Shape words	Not identified.	0.06116

	13 currently dieting.									
Green et al (1994)	N = 61 females in the UK; 20 AN, 41 CT undergraduate students	AN in out-patient Tx (EAT M 41.6), CT undergraduate students (EAT M 9.3)	AN Mbmi = 17.9, CT Mbmi = 21.2.	2 (Group: AN vs control) X 2 (Word: experimalestal vs control) ANOVA.	Two-way attenuated interaction	*F*(1, 59) = 5.61	Body shape, Food	Log transformed reaction times (milliseconds)	No clear hypothesis.	0.02115
Green et al (1999)	74 females in the UK; 34 AN, 39 CT	AN group had a clinical diagnosis (DSM-IV) and were receiving inpatient Tx. CT group were female undergraduate students.	Middle-class, Mage = 21.86. AN group Mbmi = 17.2, CT group Mbmi = 21.1.	4 (word type: body shape, birds, animals, hetergeneous) X 2 (subject group: AN, control) ANOVA	Two-way attenuated interaction	*F*(3, 213) = 4.38	Body Shape	RT (milliseconds)		0.00514

Hodge et al (2020)	N = 120 female college students in the USA. 48 women with restrained eating, 44 CT.	Elevated DEB-Q scores were used to determine restrained eating vs. non-restrained eating groups.	Mage = 19.51, 67.5% Caucasian.	2 (Group: restrained, non-restrained eaters) X 3 (Word type: forbidden food, non-forbidden food, control animal words) ANOVA	Interaction	*F*(1, 90) = 0.10	Food words: forbidden and non-forbidden & animal control words.	RT (seconds)	They also examined the 2 (Group) X 2 (word type: food ~ both forbidden and non-forbidden vs control) interaction. No sig interaction, but sig main effect of group &	0.75256

									main effect of words type.	
Jansen et al (1998)	*N* = 28 female students in the Netherlands. 15 with low restraint, 13 with high restraint.	Participants with high restrained eating *M* RS 17.6(5); Participants with non-restrained eating *M* RS 6.1(3).	Normal BMI in all participants.	4 (Word type: unmasked body, unmasked neutral, masked body, masked neutral) X 2 (Group: high, low restrained eating) X 2 (Task: subliminal, supraliminal) *ANOVA*	3-way interaction (Word type X Group X Task)	F(1,26) = 2.83	Body	RT (milliseconds)		0.10449
Kanakam et al (2013)	N = 70 female twins from the UK. 51 twins with EDs; 19 control twins	Twins with an ED were diagnosed with either BN or AN according to the DSM-IV.	91% Caucasian British. English was first language for all but one set of twins. Age Means for monozygotic twins with an ED = 38.28, without an ED = 46.64, dizygotic twins with an ED =	Two cell: twins with ED vs. twin without ED	Mean difference	*chi2*(2)=15.63	threatening (angry) faces	Attentional bias variable is derived by subtracting the mean time	Unclear hypothesis.	0.0004

without EDs	39.17, without an ED = 42.88, CT = 42.57.	to colour-name nonsocial stimuli from the mean time to colour-name all of the social stimuli. Positive scores indicate longer response latenc

								ies for social stimuli, and negative scores indicate longer response latencies for non-social stimuli. Milliseconds.			
Kuehfuss (2007)	N = 36 females in the USA. 4 with AN, 11	Participants in the ED group were receiving	Mage 24. The participants' ethnic background was 58% Caucasian, 17% Hispanic,	3 (Group: control, Obes., ED) X 3 (Food valence: positive, neutral, negative) ANOVA	Two-way attenuated interaction (Group X Food valence)	*F*(3, 26) = .74	food, body	RT (milliseconds)	Study was very underpowered	0.53781	

	with BN, 15 average weight controls, 6 controls with obesity (BMI >30).	Tx from eating disorder programs in California. The control groups were university students.	3% Asian-American, and 22% who did not specify.						(i.e., less than 30 participants in each group).	
Lattimore et al (2000)	N = 152 girls in the UK; High DFT = 35, Low DFT = 67.	Participants were grouped based on a median split on scores from the restraint subscale of the DEB-Q.	High DFT Mage 14.4; Mbi 22.7; Low DFT Mage 14.2, Mbmi 20.1.	Two cell: High DFT vs. Low DFT	Mean difference	*t*(100) = 1.55	food, body	RT (seconds)		0.1243

Lee et al (2017)	*N* = 39 females in Korea; 14 healthy controls; 13 BN 12 BED.	Participants who scored below 21 points on the EAT-26 were classified as CT. Those in the ED group were formally diagnosed through clinical interviews with a psychiatrist using DSM-5 criteria.	CT group Mage 23.3, Mbmi 20.4; group with BN Mage 23.6, Mbmi 25.6; BED Mage 23.7, Mbmi 21.5	3 (Group: CT, BN, BED) X 2 (Stimuli type: food, control) *ANOVA*	Two-way attenuated interaction (Group X Stimuli type)	*F*(2,36)=2.075	food	Interference measured by accuracy and reaction time. We used reaction time as the main value because it is more in line with other studies. This doesn	0.1432

Léonard et al (1997)	N = 92 females in France; 16 AN restricting; 25 AN binge eating ; 20 BN; 29 CT	Clinical diagnosis (DSM-IV) receiving outpatient Tx. Controls were staff and students in medicine or psychology at the mental health clinic.	AN restricting Mage 22.8,Mbmi 14.9; AN binge eating Mage 25.6, Mbmi 16.5; BN Mage 26.2, Mbmi 20.8; 29 CT Mage 25.4, Mbmi 20.2	4 (Group: AN restricting, AN bingeing, BN, control) X 2 (Target word: control, food words) ANOVA	Two-way attenuated interaction (Group X Target word)	*F*(3,88)=7.21	food, body	't change anything as both interactions are non-significant. the time taken to colour-name a complete set of 96 words using a stop watch (seconds)	Food words interaction was significant, body words was not.	0.0022

Lokken (2001)	N = 60 undergraduate females in the USA; 30 CT, 30 at risk for ED.	CT (scores below 40 on the BULIT-R), risk for ED (score above 80).	CT Mage 19.5, Mbmi 22.2; at risk for ED Mage 19.4, Mbmi 23.4. 95% caucasian, 1.7% african american, 3.3% native american. 56% under $20,000/year, 8.3% $20 - 50,00/year, 18.3% $50 - 100,000/yr, 6.7% >$100,000, other/unknown 10%.	2 (Group: control, at risk) X 2 (Target word: ED stimuli, neutral stimuli) ANOVA	Two-way attenuated interaction (Group X Target word)	*F*(1, 58) = 5.62	body shape images of models	RT (seconds)	Standard colour stroop task on top of images.	0.0211
Lokken et al (2006)	N =90 undergraduate females in the USA; 30 control, 30 subclinical ED, 30 BN	Subclinical ED group had BULIT-R scores at or above 80; CT group had BULIT-R scores at or below 40; BN	CT Mage 19.5, Mbmi 22.2; subclinical ED Mage 19.4 Mbmi 23.4; BN Mage 19.1, Mbmi 22.1	Three cell design (group: control, subclinical, BN)	Mean difference	*F*(6,170)= 5.77	food(eating), weight, shape (averaged together)	Time (seconds) to color-name words on experimental stimuli minus time to		0.00002

		group had self identified BN that fit DSM-IV criteria.						color-name words on control stimuli.	
Long et al (1994)	N = 133 females in the UK: 37 AN, 51 with obesity and restrained eating, 45 CT	Participants in the AN group were receiving inpatient Tx for DSM-III-R diagnosed AN. Participants with obesity and restrained eating were recruited from a weight loss centre.	AN group Mage 25.8, Mbmi 16.5; group with obesity and restraining eating Mage 35.8, Mbmi 26; CT Mage 42.4, Mbmi 22.5.	Four 1-way (group: AN, obes. restrained eating, control) ANOVAs split by card/word type (Control-food, Control-body, food, and body).	Main effect of group	*F*(2, 130) = 4.33	Food	Seconds	0.01 512

		CT group were occupational therapy students.							
Lovell et al (1997)	N = 109 females in the UK: 31 current AN, 11 current BN, 23 recovered AN, 11 recovered BN, 33 CT	Clinical diagnosis (DSM-III-R), 2 years or more recovered from a previous diagnosis. All groups, and controls, were recruited by notices requesting local volunteers for a	Current AN Mage 25.5, Mbmi 16.6; current BN Mage 26.9, Mbmi 21.8; recovered AN Mage 29.3, Mbmi 22.2, recovered BN Mage 34.4, Mbmi 24.6; CT Mage 24.7, Mbmi 22.9.	5 (Group: current AN, current AN, recovered AN, recovered BN, CT) X 6 (Word condition: shape words, matched control) ANOVA	Two-way attenuated interaction (Group X Word condition)	*F*(4, 116) = 5.64	Shape words	RT (Seconds)	0.00035

		health study.								
Markis (2015)	115 females in the USA; 51 receiving outpatient Tx for AN or BN; 60 controls.	All participants had a DSM-5 diagnosis of AN or BN. CT participants were recruited from a psychology undergraduate student pool.	Age range approximately 21 - 59. Primarily caucasian 73%, <1% african american, hispanic/latino, asian, european american.	2 (Group: ED, Control) X 2 (Word type: body, neutral)	Two-way attenuated interaction (Group X Word type)	F(2, 220) = 1.20	body	RT (milliseconds)		0.30316
Mcmanus et al (1996)	N = 60 females in the UK; 30 BN, 30 CT	Clinical diagnosis (DSM-IV) of BN. CT group	BN Mage 22.5, CT Mage 22.5	Two cell design (Group: BN, control)	Mean difference	Z=*.21*	threat	Interference scores were	The statistics are not presented	0.83367

was recruited from a pool of undergraduates.

calculated by subtracting time taken to color-name each control list from the time taken to color-name the threat list. A higher score reflected a greater attent

in the body of the paper. We selected the key stats from the first available line in the table.

								ional bias.		
Muller (2000)	*N* = 65 female college students in the USA.	Body shape pre-occupation (higher vs. lower) measured by the Body Shape Questionnaire.	Mage 20.5 years. Mbmi 23.6. 81.5% caucasian, 10.2% african american, 3.6% other, 2 % Asian.	Association	Correlation	*F* (1, 63)= 4.50	body	RT in tenths of a second. Errors also recorded.		0.03784
Ogden & Greville (1993)	N = 56 females in the USA; 28 participants with restrained eating,	Median split using the DEB-Q and the RS.	Restrained eaters MRS 35.6(6), unrestrained eatings MRS 17.6(5)	2 (Condition: high vs. low calorie preload) X 2 (Restraint level: Restrained vs. unrestrained eaters) ANOVA	Two-way attenuated interaction (Condition X Restrained level)	*F*(1,52)=2.69	food, body	The difference between the reaction time for the stimu	Results for time 2 were used because this was more in line with	0.10702

| | | |

						2]=4.48, p=<.05)				
Perpina et al (1993)	N = 64 females in spain; 18 AN in out-patient Tx, 14 BN in outpatient Tx, 32 CT	AN and BN had Clinical diagnosis; using Russell (1979) diagnostic criteria. Controls were recruited by sending letters to the community aking for volunteers for a color perception study.	AN Mage 26.4, Mbmi 17.2; BN Mage 26.4, 26.6; CT Mage 26.9, Mbmi 23.4.	3 (groups: AN, BN, CT) X 2 (Condition: body, control) Repeated ANOVA	Two-way attenuated interaction (Group X Condition)	*F*(1, 62) = 5.38	Body shape-size	RT (seconds)		0.02368

Perpiñá et al (1998)	N = 43 females in Spain; 15 AN, 10 BN, 18 CT.	Participants in the AN and BN groups had a clinical diagnosis (DSM) and were in outpatient Tx for either AN or BN. CT group was recruited from outpatient staff.	AN Mage 23.8, Mbmi 16.2; BN Mage 27.8, Mbmi 21.1; CT Mage 29.3, Mbmi 21.1.	3 (Group: AN, BN, control) X 2 (Condition: food, control) ANOVA	Two-way attenuated interaction (Group X Condition)	F(2,40)=2.27	food, body size	The number of words color-named per second.	I selected the first ANOVA that appeared in the paper comparing the three groups (AN, BN, CT) by cue word (target, control), because this was the initial	0.11647

									study design.	
Phelan et al (2011)	N = 48 females in the USA; 19 CT , 14 with obesity, 15 with long-term weight loss maintainers.	CT BMI of 18.5-25; Participants with obesity BMI equal to 30 or above; weight loss maintainers kept off 30lbs or more or at least 3 years. Those with obesity	CT Mage 43.6, Mbmi 21.6; group with obesity Mage 48.3, Mbmi 34.3, long-term weight loss maintainers Mage 48.5, Mbmi 23.7.	2 (Food condition: low-calorie, high calorie) X 3 (Group: those with obesity, normal weight, long-term weight loss maintainers) ANOVA	Two-way attenuated interaction (Food condition X Group)	*F*(2, 42) = 3.6	Food (low vs high calories)	RT (milli seconds)		0.03606

		had elevated ED psychopathology indicated by the EDI.								
Preuss et al (2019)	N = 101 females and males in Germany; 24 with BED, 47 with obesity non-BED, 30 CT.	Participants with BED and obesity were recruited from outpatient treatment centres. Those in the BED group met the diagnostic critera for BED (DSM-5).	87 females. 16 had secondary education, 77 had university entrance. Group with BED Mage 37.44, Mbmi 32.19; Group with obesity non-BED Mage38.05, Mbmi 33.48; CT Mage 36.30, Mbmi 23.96.	3 (Group: BED, non-BED overweight, HC) X 3 (Word type: food, body, control) ANOVA	Two-way attenuated interaction (Group X Word type)	F(2,98) = 2.426	food, body	RT (milliseconds)	Unclear Hypothesis. Non-significant interaction. If controlled for word valence the main effect of group reached significance	0.09368

									F(2, 97) = 3.108, p = .049.	
Prystanski (2011)	N = 52 females and males in Canada; 16 with Seasonal Depression ,11 with ED, 25 CT.	36 university students recruited, 16 non-students recruited from a mental health clinic and the community. Self-report measures of seasonal depression and disordered eating	18 males, 34 females. Group with Seasonal Depression Mage 28.62, group with ED Mage 38.91, CT Mage 27.80. N = 49 Caucasian, 2 Asian, 1 Other.	3 (Group: SAD, ED, control) X 2 (Word type: food, control OR body, control)	Two-way attenuated interaction (Group X Word type)	NS - not reported	food, body	the difference in response latency (ms) between a stimulus word (body shape, binge, non-binge, winter, summ		NS

		were used to classify groups. The EDDS was used for the ED group classification.						er) and its neutral word.	
Quinton (1998)	N = 100 females in the UK; 81 non-dieters, 19 dieters.	non-clinical student sample. "Dieters" were defined as women who were currently dieting in order to lose weight. The dieting group was also	Non-dieters Mage 23.75, Mbmi 21.94; dieters Mage 25.26, Mbmi 27.68. All native english-speaking Caucasian.	2 (Group: dieter, non-dieter) X 2 (Word type: threat, neutrak)	One-way Main Effect of Word Type for each group	NS - not reported	threat	A Stroop threat-word Disruption score was derived by subtracting the Stroop Neutral task time	NS

		significantly higher scoring on the EAT (M 12.26 dieting, M 3.67 non-dieting)						from the Stroop Threat task time. Seconds.	
Redgrave et al (2008)	N = 12 females in the USA; 6 AN, 6 HC	Participants with AN were receiving in-patient Tx and had a DSM-IV diagnosis. CT were recruited from universities. Groups also differed on the EAT.	AN Mage27, Mbmi = 15.7, Meducation 15 years; HC Mage 24, Mbmi = 22.3, Meducation 17 years. 10 caucasian, 2 other.	2 (Group: ED, control) X 4 (Condition: control, neutral, fat, thin)	Two-way attenuated interaction (Group X Condition)	*F*(3,8)=5.09	food and body simultaneously	RT (milliseconds)	0.02925

Ruiz et al (2008)	N = 72 females in Mexico; 36 HC; 26 BN; 10 AN.	Participants were receiving in-patient Tx for AN or BN. CT group was recruited from schools in the community and did not self-report any current or past psychiatric disorders.	*M*age 23.8 years.	Two cell design (ED, control)	Mean difference	*F*(2,69)=3.30	food [positive, negative], shape [positive,negativel]	Interference (number of errors) and response latency (milliseconds).	We have included the response latency statistics to keep with the more commonly used method. Unclear hypothesis.	0.0479

Sackville et al (1998)	N = 73 females in Australia. 53 HC, 20 AN.	HC were univeristy studnets. The HC group was split into 33 low & 20 high dietary restraint on the RS. Participants with AN were receiving in-patient Tx and had a DSM-IV diagnosis.	CT low restraint Mage 18.65; CT high restraint Mage 19.5; AN age not given.	3 (Group: control, low restraint, high restraint) x 7 (Word type: negative body shape/weight, positive body shape/weight, low caloric density food, high caloric density food, negative valenced emotion, positively valenced emotion, neutral) ANOVA	Two-way attenuated interaction (Group X Word Type)	*F*(1, 70) = 731.86	Food and shape/weight	RT (milliseconds)		0.00001
Scholar (1999)	N = 50 females in the USA. Low Disordered eating	Participants were recruited from introduction to psychology	Low BUILT-R Mage 20.82, Mbmi 24.01, 14 caucasian, 8 african-american. High BUILT-R = Mage 21.36, Mbmi 28.05, 19	2 (Race: african american, caucasian) X 2 (Disordered eating: high, low) X 3 (Stroop conditions: body, travel	Three-way attenuated interaction (Race X Disorder Eating X Stroop condition)	NS, not reported.	body shape	RT (seconds)	There were no 3-way or 2-way (ED X	NS

	behaviour 22, High Disordered eating behaviour 28	courses. Participants were classified into high or low ED psychopathology using the BUILT-R.	caucasian, 9 african american.	control, "words" control)					stroop word) interactions.	
Seddon & Waller (2000)	N=80 females in the UK.	Participant were univeristy students. They were split by high (10 or higher) vs. low score (9 or lower) on the BITE (i.e., low was the CT group).	Split by age. Younger group Mage 19.7, Mbmi 21.3; older group Mage 24.7, Mbmi 22.2.	Two cell design (high BN attitudes, low BN attitudes)	Mean difference	*t*(78) =1.57	threat: emotion [negative], emotion [positive]	RT (milliseconds)	this study was separated by young and old. Young came first chronologically so we used those results.	0.12046

									Results for older females were statistically sig. for negative t(78)= 2.60, p=.015, but not for positive words.	
Stewart & Samoluk (1997)	N = 32 females and males in Canada. 11 high RS, 10 medium RS, 11	Undergraduate psychology students participated in this study. High restraint	78% female sample. Mage = 21.8. 97% Caucasian.	3 (Group: high, medium, low retrained eaters) X 3 (Card type: food, alcohol, control) ANOVA	Two-way attenuated interaction (Group X Card type)	*F*(4, 58) = 3.87	Food, alcohol, control words	RT (seconds)		0.00745

	low RS score.	group had an RS score of 16 or higher. Medium restraint group had an RS score of 11 - 15. Low restraint group had an RS score of 0 - 10.							
Tabri & Palmer (2020)	N = 86 males and females in the USA.	Participants were recruited from MTurk. The BAAS was used as a measure of ED pscyhopathology.	39 males and 47 females, Mage 37.78.	2 (appearance overvaluation: higher vs. lower) X 2 (Word domain: appearancevs. nonappearance) × 3 (Word valence: negative, positive, neutral)	Interaction	*Z* = 3.06	body shape (stigmatized words, positive words)	RT (seconds)	0.00221

Tapper et al (2008)	N = 224 female and male undergraduates from the UK, Greece, and Iran. 164 low restraint, 28 high restraint non-dieting, 24 high restraint and dieting.	Restraint was categorized using a median split on DEB-Q scores.	Swansea, UK (36 females, 30 males), Ioannina, Greece (30 females, 30 males); Mashhad,Iran (60 females, 38 males). Mage = 21.7 years Mbmi = 22kg/m2.	One-way ANOVA (group: low restrained non-dieter, high restrained non-dieter, high restrained dieter)	Main effect of group	*F*(2, 213) = 3.02	Food	Scores were made by subtracting the overall time it took to read the neutral words from the time it took to read the food words	No clear hypothesis.	0.0509

Tressler (2008)	129 females in the USA; 63 low-risk for ED, 65 high-risk for ED.	Participants were college studnets. Risk for ED was measured using the RS and the EDI.	Average age between 19 - 20.	2 (Group: high risk, low risk) x 2 (Word type: food/body, neutral) ANOVA	Two-way attenuated interaction (Group X Word type)	*F*(7, 103) = .51	food & body words	RT (milliseconds)		0.82519
Waller & Ruddock (1995)	N= 80 females in the UK; 20 AN, 30 BN, 30 CT	The group with BN was receiving out-patient Tx. Controls were undergraduate and nursing students.	AN Mage 25, Mbmi 17.4; BN Mage 25, Mbmi 22.4; healthy controls Mage 24.2, Mbmi 21.7.	Two cell design (ED, healthy control)	Mean difference	*F*(1,78) = 6.09	food, abuse	The food-word difference score was obtained by subtracting the Food-word control task time from the Food-	this study initially had 6 groups of participants, but the difference was not significant in this case, and the statist	0.01578

word time.	ic from this findin g is not provi ded. The author s collap sed the ED group s and the contro l group s into two main group s to get the report ed findin g.

									This was deemed acceptable for inclusion because the abuse hypothesis is not relevant to this research project.	
Wilson (1999)	N= 77 females in the USA. 19 BN, 27 with restrained eating; 31 CT	Undergraduate students from midwestern university. Participants were	Age 18 - 25, weight within +/- 15% of normal weight. All participants in the highest three SES classes.	3 (Clinical group: BN, RE, HC) X 2 (Word type: target, control) ANOVA	Two-way attenuated interaction (Clinical group X Word type)	F(2,73) = .37	food, body.	interference score: time to color-name food and		0.69202

						grouped using the BUILT-R, the RS, and a structured clinical interview for a DSM-III diagnosis.	body words minus time to color-name animal words

Note. Bulimia Nervosa (BN), Anorexia Nervosa (AN), Control (CT), Bulimia Test-Revised (BUILT-R), Drive for Thinness (DFT), Dutch Eating Behaviour Questionnaire (DEB-Q), Eating Attitude Test (EAT), Body Shape Questionnaire (BSQ), Restraint Scale (RS), Beliefs about Appearance Scale (BAAS), Eating Disorders Inventory (EDI), Eating Disorder Diagnostic Scale (EDDS), Bulimic Investigatory Test, Edinburgh (BITE).